

Master's Degree in Methodology of Behavioral and Health
Sciences

Establishing External Criteria for Updating a Meta-Analysis

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Abstract

The body of research that informs evidence-based practice is so extensive that meta-analysis, as a tool for quantitative synthesis, has become a standard reference for decision-making. However, as time passes, properly conducted meta-analyses become outdated, raising the question of when it is worthwhile to publish an update. It is important, similar to the sequential rules in primary studies, to establish a priori or external criteria that help us decide when it is appropriate to update a meta-analysis, especially when its results are not statistically significant.

The objective of this study is to develop a practical tool for researchers to help them decide when to update a meta-analysis based on two independent and complementary criteria: the power and the precision of the pooled estimate. A *script* has been developed for the R statistical environment that allows one to estimate how many studies would be necessary to achieve: (a) a specified power value, and/or (b) a specified reduction in the width of the confidence interval for the combined effect size estimate. These calculations can be performed for both fixed-effect and random-effects models.

Abstract

The body of research that informs evidence-based practice is so vast that meta-analysis, as a quantitative synthesis tool, has become a standard for decision-making. However, as time passes, well-conducted meta-analyses become outdated, and one may wonder when it is worth publishing an update. It is important, similar to the sequential rules in primary studies, to establish external criteria that help us decide when it is appropriate to update a meta-analysis, especially when its results are not statistically significant.

The objective of this study is to develop a practical tool for researchers to help them decide when to update a meta-analysis. This tool is based on two independent and complementary criteria: the power and the precision of the estimate of the mean effect size. A script has been developed for the statistical environment R that allows estimating how many studies would be necessary to achieve: (a) a certain level of power, and/or (b) a specific narrowing of the confidence interval for the mean effect estimate. These calculations can be performed for both fixed-effects and random-effects models.

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1. Introduction

Meta-analysis (MA) is a methodology for the synthesis and statistical analysis of a set of results from primary studies with the aim of integrating the findings (Botella & Sánchez-Meca, 2015; Cooper et al., 2019; Glass, 1976).

It is therefore a quantitative methodology for synthesizing information, which uses a systematic review to select the primary studies of interest. Once the search for eligible cases has been conducted, the results of these studies are compiled using a common effect size (ES) metric. This allows for a point estimate of the parametric effect size, which falls within a range of plausible values, or confidence interval.

Meta-analysis has established itself as a standard technique within the field of evidence-based practice (Botella & Caperos, 2019), as it is capable of synthesizing a large number of primary studies while also minimizing the biases and limitations inherent in narrative reviews. Meta-analysis aims to bring rigor to the systematic review process by seeking to achieve the objectives of objectivity, replicability, and precision that are already required of the primary studies included in the reviews (Botella & Sánchez-Meca, 2015)

However, the meta-analysis should not be viewed solely as an endpoint in scientific research (e.g., determining whether a third-generation intervention is worth using to treat depression), but can also serve as a starting point for planning the direction of future research (Rothery et al., 2022). Continuing with the previous example, a meta-analysis could suggest which populations are worth further investigation regarding the intervention's effects.

As we have seen, research influences practice, and practice influences research. This implies a continuous growth in the body of primary studies, which can cause properly conducted meta-analyses to become outdated and, therefore, not as useful as they could be when making decisions. This raises the question of when it is appropriate to update a meta-analysis, since it stands to reason that not every update will be published. This fact raises the idea of establishing criteria that allow the methodologist to decide when to update a MA.

It is important not to base the decision on arbitrary criteria, such as whether or not statistical significance was achieved in the original meta-analysis. Basing the decision on the results may

lead to the malpractice of adding cases (in this case, studies) based on significance, which results in a progressive increase in the Type I error rate. This is the same effect that occurs when adding participants to a primary study, depending on whether the significance level has been reached or not (Botella et al., 2006).

Decisions regarding the inclusion of samples or studies in a meta-analysis for the purpose of publishing an update should be guided by criteria independent of the results themselves. There are some tentative proposals, such as that of Valentine et al. (2010), who suggest setting a specific increase in the power of meta-analytic comparisons, which allows for estimating the number of studies needed to achieve it. One of the reasons frequently cited to justify conducting a meta-analysis is that combining studies will yield higher power than each study taken individually (Cohn & Becker, 2003). This claim is evident when using a fixed-effects model, but may not always hold true when using a random-effects model (Cohn & Becker, 2003; Jackson & Turner, 2017).

Another criterion that may be of interest, given its relevance in meta-analyses, is to aim for an improvement in the precision of the combined estimate of the effect size. That is, a specific reduction in the width of the confidence interval (CI). The value of this criterion lies in the fact that decisions will be more reliable the smaller the estimation error is in the values upon which those decisions are based.

The aim of this study is to develop a useful tool for methodologists to help them decide when to update a meta-analysis based on two independent and complementary criteria: the power and precision of the pooled estimate. It may also be useful for the editorial boards of scientific journals, which must decide when it is worthwhile to publish what is essentially an update of a meta-analysis's results. Let us assume that adding a single study already yields some gain in power and precision. The question is: with how many new studies (or participants) do the improvements in power and/or precision justify publication? We have developed a *script* for the R statistical environment that allows us to estimate how many studies would be necessary to: (a) achieve a given power value, and/or (b) a given reduction in the width of the CI for the combined estimate of the effect size. The tool will allow calculations to be performed for both fixed-effect and random-effects models.

2. Statistical significance of the average effect size

Meta-analyses synthesize primary studies to arrive at their results. Since these studies are drawn from samples of varying sizes, the variance in the estimates from each study will differ from that of the others. This characteristic makes it advisable to conduct meta-analytic comparisons by weighting the studies according to their precision. The weighting method will depend on the statistical model chosen. The most commonly used models in meta-analysis are the fixed-effect model and the random-effects model. Although both models use a similar weighting method, the weighting factors differ.

The fixed-effects model (FE) assumes that all studies included in a meta-analysis estimate a single parametric or population-level treatment effect (TE). The variability observed among the TE estimators is due exclusively to sampling error, or within-study variability. Thus, the weighting of the studies is calculated as the inverse of the variance:

$$w_i^{FE} = \frac{1}{\sigma_{m(i)}^2}, \quad (1)$$

where w_i^{FE} is the weight assigned to the i-th study, and $\sigma_{m(i)}^2$ is the sampling variance of that study.

The fixed-effects (FE) model is based on an unrealistic assumption: that all studies are replicates estimating a single common treatment effect, with the only difference being the sample used. Given this limitation, the random effects (RE) model emerges, which assumes that the estimated TE constitute a random sample from a population of parametric effects. Therefore, studies not only differ in sampling error but are also estimators of different parametric TE. This implies that variability stems from two sources: within-study and between-studies. Therefore, the weighting method is

$$w_i^{RE} = \frac{1}{\sigma_{m(i)}^2 + \tau^2}, \quad (2)$$

where w_i^{RE} is the weight assigned to the i-th study according to this model, and τ^2 is the variance between studies, which is common to all studies and must be estimated using the TE of the studies included in the MA. The most appropriate method for estimating τ^2 in most The maximum likelihood estimation method is restricted (Viechtbauer, 2005), and therefore this will be the method used in the calculations in this text. Since the only difference between the two models

is the weighting factors, the calculations described here will use the fixed-effect or unspecified-model notation to provide a clearer presentation. However, all calculations (with certain exceptions) apply to both models.

The main meta-analytic result is the combined estimate of the common effect size ($\hat{\theta}$, in the EF model) or the mean of parametric effects ($\hat{\mu}_{\vartheta}$, in the EA model). The optimal estimator (unbiased and with minimum variance) of ϑ is calculated as (Hedges & Olkin, 1985)

$$\hat{\theta} = \frac{\sum w_i \hat{\vartheta}_i}{\sum w_i}, \quad (3)$$

which is the mean of the TE weighted by the weights of the studies themselves. If the TE index has an asymptotically normal sampling distribution, a confidence interval (CI) for the parametric effect can be established. The CI is calculated as:

$$\hat{\theta} \pm |z_{1-\alpha/2}| \cdot \hat{\sigma}_{\vartheta} = \left\{ \begin{matrix} T_{-}^{(sup)} \\ T_{-}^{(inf)} \end{matrix} \right\}, \quad (4)$$

where $z_{1-\alpha/2}$ is the value of the standard normal distribution corresponding to the $(1-\alpha/2)$ percentile, assuming a confidence level of $(1-\alpha) \cdot 100\%$; T_{sup} and T_{inf} are the upper and lower confidence limits and the lower standard deviation of the CI, respectively; and $\hat{\sigma}_{\vartheta}$ is the estimate of the standard error of $\hat{\theta}$, which is calculated as

$$\hat{\sigma}_{\vartheta} = \frac{1}{\sqrt{\sum w_i}}. \quad (5)$$

The CI allows us to test the null hypothesis that the parametric TE is zero by assessing whether the value 0 lies within that confidence interval. The null hypothesis can also be tested using the test statistic

$$z = \frac{\hat{\vartheta}}{\hat{\sigma}_{\vartheta}}, \quad (6)$$

which, under the null hypothesis, follows a standard normal distribution. Statistical software packages provide a critical probability level (p) associated with the value of z , such that the null hypothesis will be rejected if $p < \alpha$.

3. Updating a Meta-Analysis: Power-Based Rules

The power of a test is the probability of rejecting the null hypothesis (H_0) when it is false. Since H_0 and H_1 are exhaustive, such a rejection implies the indirect affirmation of H_1 . This probability is determined by the significance level (α), set a priori; the alternative parametric effect size for which the power is to be determined (θ_1); and the standard error of the test statistic ($\hat{\sigma}$). Power depends, first and foremost, on the distribution of the null hypothesis (H_0); in most meta-analytic tests, it is assumed that, under *the true* H_0 , the test statistic is approximately normally distributed with a mean of 0 and a variance of 1. For the test to be significant, a minimum combined effect size, or critical value ($TE_{\bullet(c)}$), must be reached, which corresponds to the *z-score* at the confidence level ($1 - \alpha$). In a right-tailed one-sided test, $TE_{\bullet(c)}$ is expressed as

$$TE_{\bullet(c)} = z_{(1-\alpha)} \cdot \hat{\sigma}_{\theta_1} . \quad (7)$$

The power (*pot*) with respect to an alternative value θ_1 is, therefore, the area of the curve under H_1 that the value of $TE_{\bullet(c)}$ associated with $z_{(1-\alpha)}$ leaves to its right (Hedges & Pigott, 2001). It is calculated as:

$$pot = 1 - \Phi \left(\frac{TE_{\bullet(c)} - \theta_1}{\hat{\sigma}_{\theta_1}} \right) . \quad (8)$$

The standard error is inversely related to the number of studies, since the larger the sample size (number of studies), the greater the precision of the estimate, which implies a reduction in variance.

3.1. Number of additional studies required to achieve a given power

Given a meta-analytic contrast with a specified $TE_{\bullet}$ and power, the question arises as to how many studies must be added to achieve a higher, pre-specified power for a specific alternative value of the parameter. Let k denote the number of studies in the meta-analysis and j the number of studies that must be added to achieve the desired power. Thus, the new meta-analysis will have $(k + j)$ studies. Higher power will require a lower standard error, which must be derived from formula (8) as follows.

Power is simply the complementary probability of committing a Type II error (β), so it can be expressed as $1 - \beta$. It is the probability that the null hypothesis

($H_0: \vartheta = \vartheta_0$) is significant when the parameter value is a specific value, other than zero,

which we will denote by ϑ_1 . We will also denote by $\hat{\sigma}_{\vartheta}$ the standard error at which the desired power is achieved, and by $TE_{\bullet(c)}$ the smallest combined effect size at which the null hypothesis test would be significant in a right-tailed one-sided test. In these terms, the power is calculated as

$$1 - \beta = 1 - \Phi \left(\frac{TE_{\bullet(c)} - \vartheta_1}{\hat{\sigma}_{\vartheta}} \right) \quad (9)$$

The standard error cannot be directly derived from equation (9), since the $TE_{\bullet(c)}$ is also unknown. However, this value can be derived from the null hypothesis test from which the standardised critical value is calculated for a confidence level of $1 - \alpha$:

$$\Phi^{-1}(1 - \alpha) = z_{1-\alpha} = \frac{TE_{\bullet(c)} - \vartheta_0}{\hat{\sigma}_{\vartheta}} \quad (10)$$

By solving for $TE_{\bullet(c)}$ in the previous equation, we can establish that for $H_0: \vartheta = 0$

$$TE_{\bullet(c)} = z_{1-\alpha} \cdot \hat{\sigma}_{\vartheta} \quad (11)$$

Therefore, if we substitute the power $TE_{\bullet(c)}$ into equation (9), we can express the value as

$$1 - \beta = 1 - \Phi \left(\frac{z_{1-\alpha} \cdot \hat{\sigma}_{\vartheta} - \vartheta_1}{\hat{\sigma}_{\vartheta}} \right) \quad (12)$$

from which we can derive the standard error and express it as:

$$\hat{\sigma}_{\vartheta} = \frac{\vartheta_1}{\frac{z}{1-\alpha} - \frac{z}{\beta}} \quad (13)$$

where ϑ_1 remains the alternative parametric effect size to 0, $z_{1-\alpha}$ corresponds to the value on the standard normal curve with a cumulative probability of $1 - \alpha$, and z_{β} corresponds to the value on the same curve with a cumulative probability equal to the probability of committing a Type II error ($\beta = 1 - pot$).

The square of the standard error (variance) is equal to the inverse of the sum of the weights (both those from the original meta-analysis and those from the studies to be added). The sum of the weights (w_{k+j}) is, therefore,

$$W_{(k+j)} = \frac{1}{\hat{\sigma}_{(k+j)}^2} . \quad (14)$$

The total weight is broken down into the sum of the original weights (w_k) plus the weight of the studies to be added (w_j). Therefore, the weight that must be added to achieve the a priori determined power is obtained as

$$W_j = W - W_k . \quad (15)$$

This weight can be expressed in terms of studies in various ways. We will do so in two ways: (a) as a single study with the total weight (which allows estimating the number of participants, but only under a fixed-effects model) and (b) as several studies of sizes similar to those already included in the meta-analysis (valid for both models).

The first way to express this is as a single study, which in many cases will be very large and likely unrealistic in many fields of study, but which allows us to express the result in terms of the number of participants to be added. Specifically, to achieve a given power, we must add a single study with a variance equal to

$$\hat{\sigma}_i^2 = \frac{1}{W_j} . \quad (16)$$

Under a random-effects model, this variance would continue to be decomposed into sampling variance and between-study variance, which is constant across all studies in the meta-analysis ($\tau^2 + j$). Since the variance of this single study is presumably very small, it may be smaller than the between-study variance, leading to a theoretical impossibility. Therefore, in the vast majority of cases, this way of expressing the weight of the j added studies is only possible if a fixed-effects model is used, since in this case all variability is due to sampling variance.

By substituting the effect size in question into the formula for the sample variance, we can calculate the corresponding sample size. Table 1 lists the

formulas for the two effect size indices most commonly used in psychology: the standardized mean difference and Fisher's transformation for Pearson correlations.

Table 1

Formulas for estimating sample size using sample variance

Effect Size Index	Simplified formula	Note
d	$N = \frac{4}{\sigma^2} + \frac{2d^2}{4\sigma_m^2} \quad (17)$	It is assumed that the group sizes are equal ($n_1 = n_2$)
z_r	$N = \frac{1}{m} + 3 \quad (18)$	

The second way to express this is as the number of studies that need to be added to achieve the desired power. In reality, it is unrealistic to think that a single study would be published with a variance small enough to reduce the standard error of the combined TE estimate so significantly, especially since it would only be valid for a single EF model. To calculate a reasonable number of studies needed to achieve w_j , one can assume that the sample sizes of the studies already conducted are representative of studies in that field and, therefore, future studies can be expected to have similar sample sizes. In summary, it is assumed that each study to be added has a weight equivalent to the average weight (w) of the original studies:

$$w = \frac{W_k}{k}, \quad (19)$$

Therefore, the number of studies to be added is approximately

$$j \approx \frac{W_j}{\bar{w}}. \quad (20)$$

One could also estimate the sample size of each added study by extracting the sample variance from the total variance of each study:

$$\sigma_{m(w)}^2 = \sigma_w^2 - \tau^2. \quad (21)$$

Recall that under a fixed-effects model, the between-study variance is zero ($\tau^2 = 0$). Once the value of this variance is obtained, the sample size for each study is calculated using the formulas listed in Table 1.

Numerical example. To illustrate the points made so far with an example, we have used data from the meta-analysis by Velthorst et al. (2015). This meta-analysis includes 30 studies evaluating the effect of cognitive-behavioral therapy on the negative symptoms of schizophrenia. We selected the 28 studies in which the treatment of these symptoms was a secondary objective of the therapy. The effect size metric they provide is the difference in

standardized means (d), yielding an $ET_{\bullet} = 0.093$ and a $\sigma^{\wedge}(ET_{\bullet}) = 0.0758$. As can be

see, the test yields a non-significant result, since substituting into the formula

(6) We obtain $0.093 / 0.0758 = 1.227$, which is less than the critical value of 1.645 ($\alpha = 0.05$). We will calculate the power for a “small” effect size according to the classical taxonomy of Cohen (1992) ($\delta = 0.2$), setting a significance level of $\alpha = 0.05$. We will follow a random effects model, with the estimate of the between-studies variance $\tau^2 = 0.0962$.

The minimum effect size required to achieve statistical significance and reject H_0 in a one-tailed right-tailed test is

$$TE_{\bullet(c)} = 1.645 \times 0.0758 = 0.12469 . \quad (22)$$

Therefore, the minimum effect size required to achieve statistical significance in this case is 0.12469. The power of the test with respect to the alternative value of $TE(\delta_1) = 0.2$ is, therefore,

$$p = 1 - \Phi\left(\frac{0.12469 - 0.2}{0.0758}\right) = 1 - \Phi(-0.99) = 0.84. \quad (23)$$

The question now arises as to what value of $\sigma^{\wedge}$ will be needed to achieve a , for example, equal to 0.90 with respect to the same alternative parametric TE . The standard error required to achieve the new power ($\sigma^{\boxtimes}$) is equal to (recall that 1.645 and -1.28 are the values of the standardized normal distribution with cumulative probabilities equal to 0.95 and 0.10, respectively)

$$\sigma^{\boxtimes} = \frac{0.2}{1.645 - (-1.28)} = 0.06838 . \quad (24)$$

The sum of the weights is derived from the inverse of the square of the standard error, which is equal to

to

$$w_{k+j} = \frac{1}{0.06838^2} = 213.866 . \quad (25)$$

We now calculate the sum of the weights using the initial studies: $W = 1 / 0.0758^2 = 174.045$. Therefore
 Therefore, we can conclude that the weight to be added is equal to $213,866 - 174,045 = 39,821$. Let's
 now see how to express this result in terms of participants and in terms of studies. In terms of
 participants, this would mean adding a single study with a variance equal to
 to

$$\sigma_j^2 = \frac{1}{39.821} = 0.0251 . \quad (26)$$

Since the variance in this study is smaller than the between-study variance in the meta-analysis, it is
 theoretically impossible to conduct a primary study with this variance. For educational purposes, we
 will assume a fixed-effects model solely for the purpose of performing this calculation. Therefore, to
 determine the sample size corresponding to that variance, it must be solved for from the formula for
 the sampling variance of the standardized mean difference (Table 1). We will denote by n_1
 the size of one sample and n_2 the size of the other sample, with the total sample size equal to
 $N = n_1 + n_2$. Assuming that the sizes of each group are equal ($n_1 = n_2$), the variance for $\delta = 0.2$
 would require a sample size approximately equal to

$$N = \frac{4}{\text{var}} + \frac{2\delta^2}{4 \text{ var}} = \frac{4}{0.0251} + \frac{2 \cdot 0.2^2}{4 \cdot 0.0251} = 160.159 \approx 161. \quad (27)$$

Returning to the original random-effects model, we will now express the weight in terms of
 studies. Specifically, we will use the number of studies with a weight equal to the average of the
 original weights required to achieve w_j . The average weight of the original studies in Velthorst's
 meta-analysis is calculated as the inverse of the original standard error squared (sum of original
 weights) divided by the number of original studies

$$w = \frac{1}{0.0758^2} = \frac{174.045}{28} = 6.216 . \quad (28)$$

Therefore, the estimate of the minimum number of studies to be added to achieve a power of 0.9
 is

$$j = \frac{213.866 - 174.045}{6.216} = \frac{39.821}{6.216} = 6.406 \approx 7. \quad (29)$$

If we wanted to determine the sample size for the j studies, we must first calculate how much of the
 variance is due to sampling:

$$\hat{\sigma}_{m(w)}^2 = \frac{1}{6.216} - 0.0962 = 0.0647 . \quad (30)$$

To determine the sample size, this value must be substituted into equation (17), so the sample size of the added studies is approximately equal to

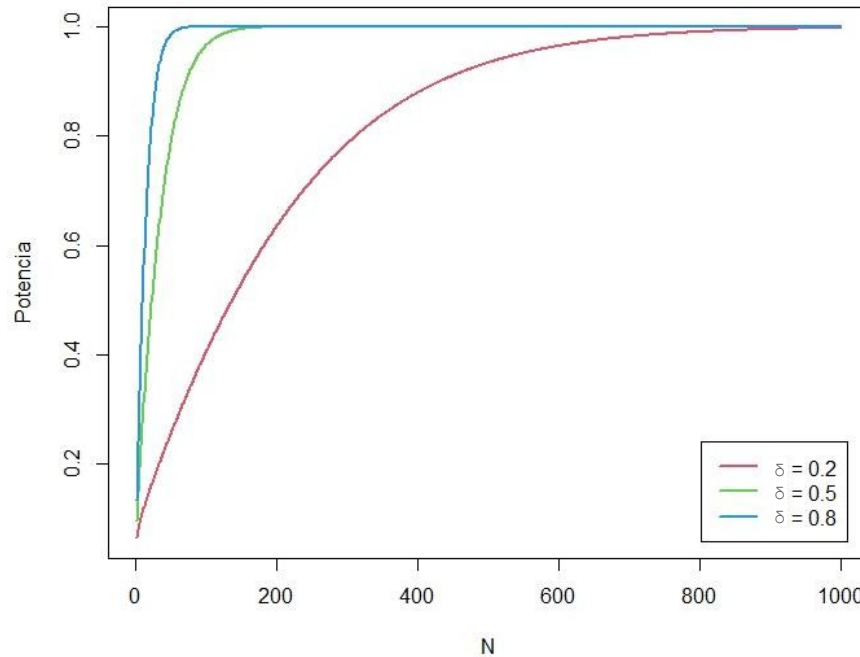
$$N = \frac{4}{0.0647} + \frac{2 \cdot 0.2^2}{4 \cdot 0.0647} = 62.133 \approx 64 . \quad (31)$$

In terms of our objective, we can conclude that to achieve the power we have established as justifying the publication of the meta-analysis ($pot = 0.90$), we would need 7 studies of average size equal to the average size of the studies already published ($N = 64$).

Fixed-effect and random-effects models. Under a fixed-effect model, adding studies to a meta-analysis will always result in a reduction in the standard error, since the weight of a study depends solely on the sampling variance (Equation (1)). Adding studies under this model, even if they have a very small sample size, will necessarily lead to an increase in the magnitude of the sum of weights. Figure 1 shows, under a fixed-effects model, how the power of a standardized mean difference (SMD) test increases as cases are added.

Figure 1

Power of a standardized mean difference test according to sample size for various alternative TE



Note. The formula for sampling variance has been used, assuming equal group sizes.

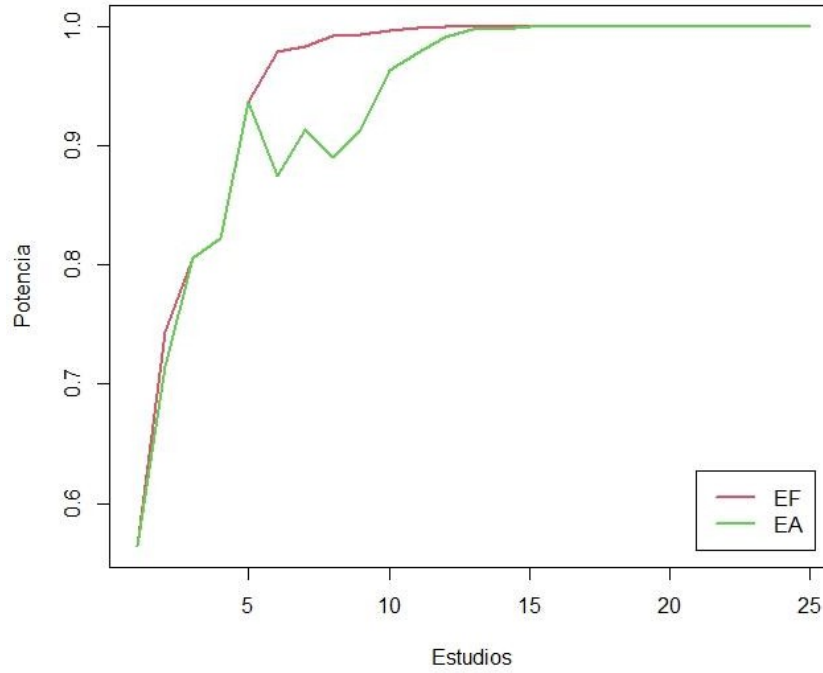
However, if a random-effects model is used, the weight of the studies depends on both the sampling variance and the between-study variance (τ^2). This variance is estimated based on the variability of the studies included in the meta-analysis; therefore, its estimate may vary if studies with a treatment effect and/or variance that differs significantly from that of the other studies are included. Re-estimating the between-study variance may result in a decrease in the meta-analysis's power as studies are added.

Data from 25 primary studies were simulated, with treatment effects (DMTs) randomly varying between -0.2 and 0.5 and sample sizes ranging from 10 to 200 (assuming equal group sizes). Figure 2 shows how the power value (for $\delta = 0.2$) changes as studies are added. The maroon line follows a fixed-effects model, while the green line follows a random-effects model, whose τ^2 has been re-estimated each time a study was added (using the restricted maximum likelihood method). It can be observed that, if a fixed-effects model is followed, the power always increases (although the magnitude of the increase

(varies), but if a random-effects model is used, the contrast power decreases.

Figure 2

Comparison of the increase in power between EF and EA in a simulated case



4. Updating a meta-analysis: Rules based on the precision of the estimate

As mentioned in the introduction, the meta-analysis technique provides both a combined point estimate of the effect size and a range of plausible values where the parametric ES (with a fixed-effects model) or the mean of parametric ESs (with a random-effects model) might be found. This range is known as the confidence interval (CI), and its width is a measure of the precision of the meta-analysis estimate. Using formula (4) as a reference, the width (a) is calculated as the difference between the upper limit of the effect size (TE_{sup}) and the lower limit (TE_{inf}):

$$a = TE_{sup} - TE_{inf} \quad (32)$$

Since the CI limits are equidistant from the point estimate of the TE (at least when an asymptotic normal distribution is assumed), the width can be expressed as twice the standard error multiplied by the value of the standard normal distribution corresponding to the percentile

$(1-\alpha / 2)$:

$$a = 2 \cdot \left(|z_{1-\alpha/2}| \cdot \hat{\sigma}_{\vartheta} \right) . \quad (33)$$

Thus, the calculation of the amplitude is independent of the effect size.

Just as with power, the standard error is inversely proportional to the number of studies, since the larger the sample size (number of studies), the greater the precision of the estimate and, consequently, the narrower the CI.

Since the width is inversely related to the precision of the estimate, it is counterintuitive to think that an improvement is achieved by reducing the width. In this case, the inverse of the width can be used as a positive measure of precision.

4.1. Number of additional studies required to achieve a given width

Given a meta-analytic estimate with an estimated standard error and a specified precision, the question arises as to how many studies must be added to narrow the confidence interval and, thereby, increase the precision of the estimate to a specified value. We again denote k as the number of studies in the initial meta-analysis and j as the number of studies that must be added to achieve the desired precision. Thus, the new meta-analysis will have $(k + j)$ studies. A smaller width will require a smaller standard error, which is derived from formula (33) as

$$\hat{\sigma}_{(k+j)} = \frac{\boxtimes'}{2 \cdot |z_{1-\alpha/2}|} , \quad (34)$$

where $\boxtimes'$ corresponds to the desired amplitude, and $z_{1-\alpha/2}$ is the value of the standardized normal distribution corresponding to the $(1-\alpha / 2)$ percentile.

Once the new standard error containing the $k + j$ studies has been obtained, the procedure for calculating the number of studies needed to achieve it is the same as that described for power using formula (14).

Numerical example. We return to the meta-analysis example by Velthorst et al. (2015) to illustrate the criterion based on range as a measure of precision. Setting $\alpha = 0.05$, the range of the combined estimate of TE is

$$a = 2 \cdot (1.96 \cdot 0.0758) = 0.297 . \quad (35)$$

Suppose now that the researchers wish to update the meta-analysis when the number of new studies allows them to achieve, for example, a effect size of approximately half ($\delta' = 0.15$). The new effect size will require a lower standard error ($\sigma_{(k+j)}^{\wedge}$), which is calculated as

$$\sigma_{(k+j)}^{\wedge} = \frac{0.15^2}{1.96} = 0.0383 . \quad (36)$$

Recall that the value of the standardized normal distribution corresponding to the percentile $(1 - 0.05 / 2)$ is equal to 1.96.

As with the power-based criterion, the sum of the weights is derived from the inverse of the square of the standard error, which is equal to

$$w_{k+j} = \frac{1}{0.0383^2} = 681,714 . \quad (37)$$

The weight due to the studies to be added is equal to $681,714 - 174,045 = 507,669$. Just as we did with the power criterion, we will express this weight in terms of participants (exceptionally assuming an EF model) and in terms of studies.

In terms of participants, a single study with a variance equal to

$$\text{var}_j = \frac{1}{507,669} = 0.00197 . \quad (38)$$

As with the example of power, the variance of the hypothetical study is smaller than the between-study variance of the meta-analysis, so it is impossible to conduct a study with that variance. If we assume, for educational purposes, a fixed-effect model, the total variance is due to sampling variance. Therefore, to calculate the sample size in this meta-analysis, it must be derived from the formula for the sampling variance for the *d-index*. To do so,

we must establish an effect size which, to continue the example, will be $\delta = 0.2$.

We will denote the size of one sample by n_1 and that of the other sample by n_2 , where the size

The total sample size is equal to $N = n_1 + n_2$. Assuming that the sizes of each group are equal ($n_1 = n_2$), the sample size is approximately equal to

$$N = \frac{4}{0.00197} + \frac{2 \cdot 0.2^2}{4 \cdot 0.00197} = 2040.61 \approx 2042 . \quad (39)$$

Therefore, a study with a total sample size of 2042 subjects would be required; that is, two groups of size $2042 / 2 = 1021$ participants.

We return to the original EA model to express the weight in terms of studies. Specifically, the number of studies with a weight equal to the average of the original weights required to achieve w_j . The average weight of the original meta-analysis studies by Velthorst $w = 6.216$, as calculated in equation (28). Therefore, the estimated studies to be added to reach a power of 0.15 is

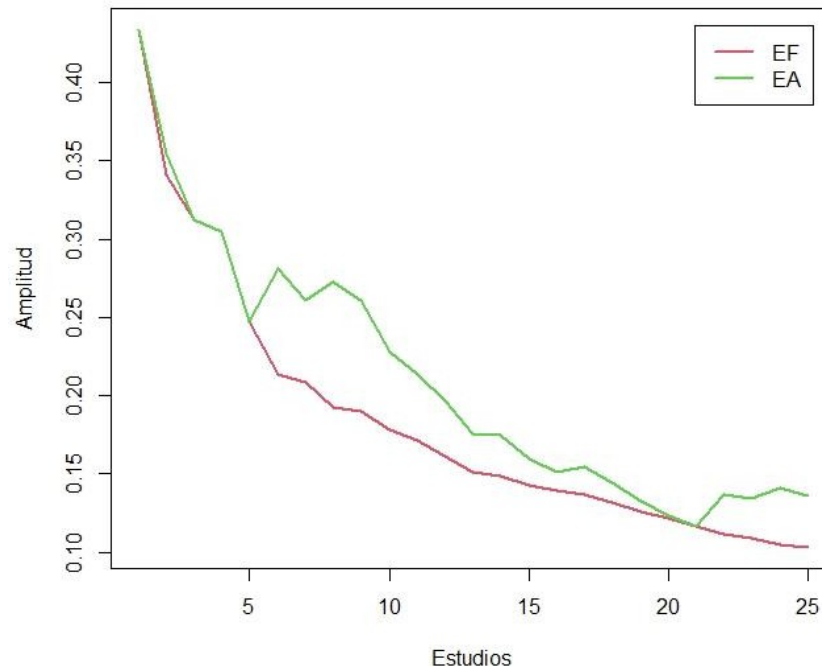
$$j = \frac{507.669}{6.216} = 81.671 \approx 82 . \quad (40)$$

It can be concluded that for the confidence interval width to reach the value we have established as justifying an update of the meta-analysis (0.15), 82 studies of the same type as those included in the original meta-analysis must be added, with an average sample size also equal to that of the studies ($N = 64$ for a $d = 0.2$).

Fixed-effect and random-effects models. As with statistical power, pooling studies under a fixed-effects model—regardless of their variability—always results in a narrower confidence interval for the pooled effect size estimate. However, under a random-effects model, the estimate of the between-study variance may increase if it is re-estimated after aggregating studies with high variability, which may lead to an increase in the width. Figure 3 illustrates how the width decreases when following a fixed-effects model and a random-effects model by re-estimating τ^2 after adding each study. The data used are the same as those simulated for the creation of Figure 2.

Figure 3

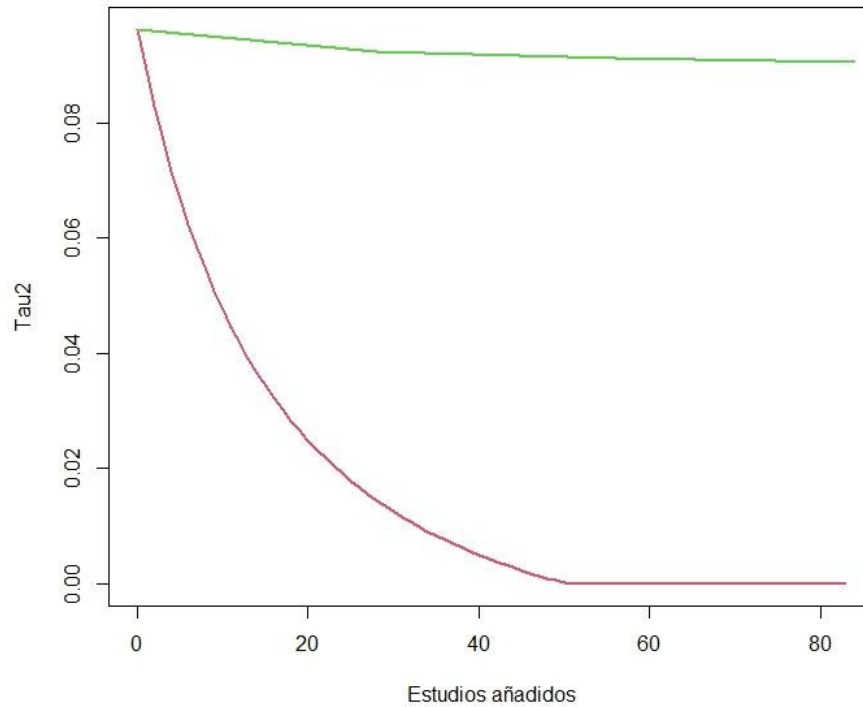
Comparison of the reduction in IC amplitude between EF and EA in a simulated case



For the calculations presented in this text, it is assumed that the data from the original meta-analyses are a representative sample of the study population; therefore, the estimate of τ^2 should be reliable. Furthermore, our estimation procedure assumes that the studies to be added are similar to those already included in the original meta-analysis; thus, in the long run, the between-study variance should remain constant. If we were to re-estimate τ^2 with each “similar” study added, it would decrease drastically and artificially, biasing the estimate. To illustrate this with an example, Figure 4 shows how the between-study variance of the meta-analysis by Velthorst et al. (2015) changes when re-estimated after adding similar studies using our procedure (with a TE and weight equal to the mean of the studies originally included in the meta-analysis) and when re-estimated by doubling, tripling, and quadrupling the original studies. Both procedures involve the long-term aggregation of very similar studies, but τ^2 is drastically reduced when re-estimated using our procedure. Therefore, to avoid artificially skewing the between-study variance, we keep this value constant after aggregating the j additional studies needed to achieve a given power.

Figure 4

Change in τ^2 when re-estimated by adding studies



Note. The change in τ^2 when re-estimated after each “similar study” is represented by the maroon line, while the change when doubling, tripling, and quadrupling the original studies is represented by the green line.

If a meta-analysis exhibits high variability, and there are sufficient reasons to believe that the estimate of between-study variance may be inadequate, we do not recommend using the criterion outlined here to decide when to publish an update of that meta-analysis.

4.2. The Hartung and Knapp Method

The standard method for performing meta-analytic tests assumes a normal distribution, with the test statistic being z (Formula (6)). However, if a random-effects model is used, the uncertainty regarding the variability caused by the estimate of the between-study variance (τ^2) is not taken into account. This results in the possibility

that the test statistics may have higher Type I error rates than the nominal significance level (α).

To address this problem, Hartung and Knapp (2001a, 2001b; see also Hartung, 1999; Sidik & Jonkman, 2002, 2006) proposed an improved estimator of the standard error of the pooled estimate, assuming a Student's *t-distribution* with $k - 1$ degrees of freedom. Since the degrees of freedom depend on the included studies, an iterative calculation process would be necessary to estimate how many additional studies would be needed to achieve both a power and a magnitude set a priori.

5. An R script to analyze and predict power and effect size

We have developed a script for the R statistical environment that allows the researcher to perform the calculations described here in a straightforward manner (see Appendix 1). R is a highly versatile open-source *software* that allows users to program all the calculations and processes they wish to perform, as well as to use code developed by other users, which is typically compiled into “packages.”

Pre-programmed calculations are typically executed as functions, which operate on a series of *inputs* or “arguments” (in R terminology) and produce one or more *outputs*. Outputs can be stored as “objects” so that they can later be manipulated more easily.

To replicate the calculations presented here, we have programmed four functions, all related to the difference in standardized means:

- “*power.ma*”. Calculates the power of a right-tailed one-sided test of the combined estimate of a meta-analysis for various alternative values of effect size (in addition to those entered manually, it always calculates the power for the classic effect size cutoff points proposed by Cohen: 0.2, 0.5, 0.8).
- “*range.CI*”. Calculates the width of the confidence interval for the pooled estimate of a meta-analysis.
- “*studies.power*”. Calculates how many studies similar to the original ones would be needed to achieve a pre-specified power for various alternative effect size values (in addition to those entered manually,

always calculates for the classic cut-off points proposed by Cohen: 0.2, 0.5, 0.8). It also performs these calculations in terms of sample size for standardized mean differences (assuming equal group sizes; only for EF). The outputs are these values, along with a graph estimating the power the meta-analysis would achieve by adding studies one by one (marking with a vertical line the number of studies needed for the power entered as *input*).

- “*studies.range*”. Calculates how many studies similar to the original ones would be needed to achieve a pre-specified confidence interval width for the pooled estimate. It also performs these calculations in terms of sample size for standardized mean differences (assuming equal group sizes and entering a TE value as *input*; for EF only). The outputs are these two values, along with a graph estimating the width the confidence interval would reach by adding studies one by one (marking with a vertical line the number of studies needed for the width entered as *input*).

The code includes numerous comments (any text written after a “#” symbol) that allow the user to effectively utilize all the developed functions. The general outline of the code is as follows:

- Function name and description
- Arguments required to execute the function and, if necessary, the possible values they can take.
- Function code

Some of the code’s calculations are performed using the “metafor” package (Viechtbauer, 2010), so it must be loaded into the workspace for the developed functions to work correctly. If the package has never been installed, it must be installed using the command: `install.packages("metafor")`.

Example of use. To help the reader become familiar with the functions, we will replicate the calculations previously performed for the meta-analysis by Velthorst et al. (2015). First, we load the meta-analysis data (included in Appendix 2) as a *data frame*:

```
data.velthorst <- read.table("velthorst.csv", header = T, sep = ";")
```

The *data frame* consists of 28 observations (the number of studies) and three variables: a number from 1 to 28 as the study identifier (n), the effect size of each study (ES), and its sample variance (var). Next, we load the “metafor” package into the system using the command:

```
library(metafor)
```

Next, we run the rest of the code to load the four functions into the global environment. Once loaded, we run the first function to determine the power of the contrast. As arguments, we specify (1) the column of *the data frame* containing the effect sizes (in our case, ES); (2) the column of *the data frame* containing the study variances (var); (3) the name of *the data frame* containing the above variables (data.velthorst); (4) the method for estimating τ^2 or whether a fixed-effects model is used (restricted maximum likelihood, “REML”); (5) the alternative parametric effect size against which the power will be calculated (0.2); and (6) the significance level (0.05).

```
power.ma(ES = ES, var = var, data = data.velthorst, method = "REML", theta = 0.2,  
         alpha = 0.05)
```

The function returns the following result:

	ES	Power
1	0.2	0.8400466
2	0.5	0.9999996
3	0.8	1.0000000

To calculate how many studies similar to those already included are needed to achieve a power of 0.90, we run the “studies.power” function with the following arguments: (1) the column in *the data frame* containing the effect sizes (ES); (2) the column in the *data frame* containing the variances for each study (var); (3) the name of the *data frame* containing the previous columns (data.velthorst); (4) the method for estimating τ^2 or whether a fixed-effect model is used (restricted maximum likelihood, “REML”); (5) the alternative parametric effect size against which the power will be calculated (0.2); (6) the significance level (0.05); and (7) the desired power (0.9). You can also include the number of studies used to generate the plot, but in this case we will leave it at the default value (50).

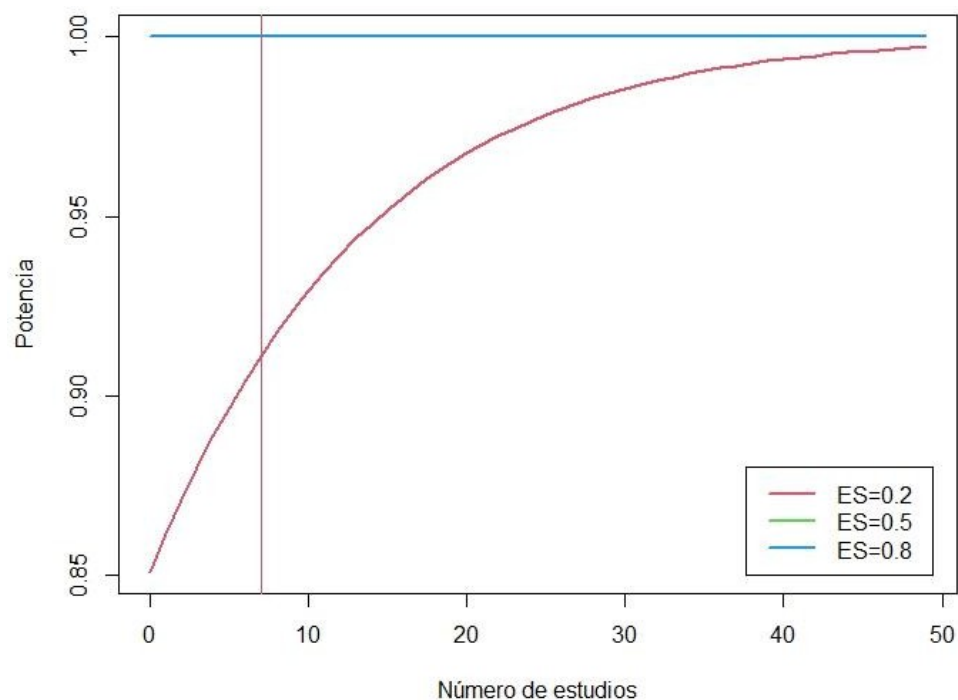

```
studies.power(ES = ES, var = var, data = data.velthorst, method = "REML", theta
              = 0.2, alpha = 0.05, power = 0.9)
```

The results are shown in Figure 5, along with the number of studies similar to the original ones estimated to achieve the specified power (`$Studies`) and the data used for the plot (`$Data.plot`), which are shown in Appendix 3. If the desired power has already been achieved with the original studies, the function returns the value 0 (since no additional studies are needed).

```
$Studies
  EN Studies
1 0.2      7
2 0.5      0
3 0.8      0
```

Figure 5

Increase in the power of the meta-analysis by Velthorst et al. (2015) based on the number of studies added



To calculate the confidence interval width, we run the “`range.CI`” function with the following arguments: (1) the column of *the data frame* containing the effect sizes (ES); (2) the column of *the data frame* containing the variances of each study (var); (3) the name of *the data frame* that

contains the previous columns (data.velthorst); (4) the method for estimating τ^2 or whether a fixed-effect model is used (restricted maximum likelihood, “REML”); and (5) the significance level (0.05).

```
range.CI(ES = ES, var = var, data = data.velthorst, method = "REML", alpha = 0.05)
```

The function returns the following result:

```
[1] 0.2970202
```

To calculate how many studies similar to those already included are needed to reduce the range to 0.15, we run the “studies.range” function with the following arguments: (1) the column in the *data frame* containing the effect sizes (ES); (2) the column in the *data frame* containing the variances for each study (var); (3) the name of the *data frame* containing the previous columns (data.velthorst); (4) the method for estimating τ^2 or whether a fixed-effect model is used (restricted maximum likelihood, “REML”); (5) the significance level (0.05); (6) the desired effect size (0.15); and (7) the number of studies to include in the plot (since 50 studies will not be sufficient to reach the desired value, we expand it to 100).

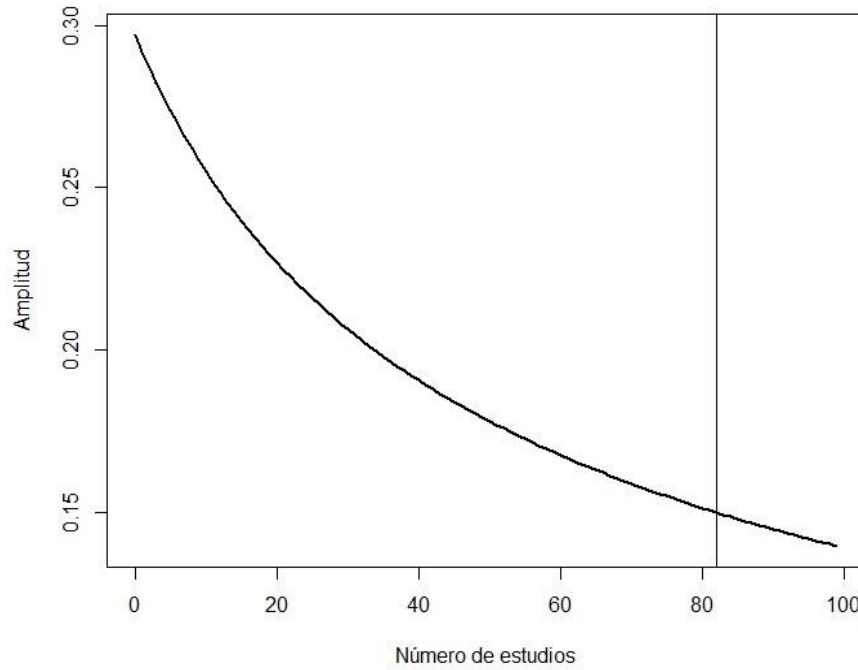
```
studies.range(ES = ES, var = var, data = data.velthorst, method = "REML", alpha  
= 0.05, range = 0.15, plot.limit = 100)
```

The results are shown in Figure 6, along with the number of studies similar to the estimated originals (\$Studies) and the data used for the plot (\$Data.plot), which are shown in Appendix 4.

```
$Studies  
[1] 82
```

Figure 6

Reduction in the CI width of the meta-analysis by Velthorst et al. (2015) according to the number of studies added



As can be seen, all values calculated using the *script* have been replicated (with minor differences due to rounding), allowing us to reach the same conclusion as if the calculations were performed manually.

6. Discussion and future research directions

There are various reasons for publishing a meta-analysis on a topic for which another meta-analytic review already exists. It may be that we disagree with the decisions made in the first meta-analysis, which would justify a re-analysis. However, in this study we have focused on a situation that is becoming increasingly common: the emergence of new primary studies that cast doubt on the validity of a properly conducted meta-analysis.

This study serves as an initial approach to establishing external criteria to guide the updating of a meta-analysis. The criteria developed in this paper are offered as a possible guide for updating a meta-analysis when the goal is to improve either the power or the width of the confidence interval (CI) for the relative risk estimate

of the TE. However, there are other meta-analytic contrasts that can serve as a reference when deciding when to update a meta-analysis. One such contrast is the analysis of moderators, whether categorical or continuous. Furthermore, the power of these contrasts is typically much lower than that of the combined estimate (Valentine et al., 2010). A future line of research should focus on this area, developing the necessary calculations to estimate how many studies are needed to improve the power of the moderator contrast.

Another aspect to explore is the sensitivity or impact that the use of these criteria has on the publication bias inherent in meta-analyses. This bias refers to the higher likelihood of including statistically significant studies in meta-analyses, to the detriment of potential unpublished research that does not yield significant results or, failing that, yields very small effect sizes (Botella & Sánchez-Meca, 2015). There are many reasons why publication bias may arise, such as self-censorship or the difficulty of publishing non-significant results in scientific journals (Francis, 2012).

The consequence of publication bias is an unrealistic combined estimate of the effect size, as it is positively biased. Regarding the guidelines presented here, we could tentatively assume that they would exacerbate the impact of publication bias, minimizing the number of studies needed to achieve the desired power or precision. This is because the “similar studies” used for the estimates would also be positively biased by publication bias. However, one of the cases in which the application of these criteria (especially that of power) would be of greatest interest is the updating of non-significant meta-analyses, which presumably suffer less from publication bias.

Finally, another line of work should focus on improving and refining the R *script*, adding new features such as the calculation of standard error using the Hartung and Knapp method, the calculation of other effect size indices, as well as error control structures. Another objective is to develop the code as a package so that it can be easily used by any R user.

7. References

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8. Appendices

8.1. Appendix 1. R script

```
## Required support packages ##
# install.packages("metafor") # If the package is not installed library(metafor) #
Used to calculate the variance of the combined estimate

## -power.ma-          Function for calculating the power for a right-tailed one-
tailed test ##

# Arguments #
# -ES-                Column of a data frame containing the effect sizes for each
study (For metafor)
# -var-                Column of a data frame containing the variances for each study
(For metafor)
# -data-              Data frame containing the data (effect sizes and variances) (For
metafor)
# -method-            Method for estimating tau^2 (For metafor). The most common codes
are (run ?rma.uni for all codes):
#   -"FE"-            (Fixed effect. Tau^2 is not estimated)
#   -"REML"-          (Restricted Maximum Likelihood). By default
# -theta-              Alternative parametric TE. By default, the power is calculated for the
classic cut-off points proposed by Cohen (0.2, 0.5, 0.8), and these higher values
are always offered.
# -alpha-              Significance level. Default = 0.05

# Function implementation #
power.ma <- function(ES, var, data, method="REML", theta=NULL, alpha=0.05) {

  # Vector of sorted alternative effect sizes ES <-
  sort(c(theta, 0.2, 0.5, 0.8))
  ES <- ES[!duplicated(ES)] # Remove duplicates if any

  # Calculate the standard error
  ma <- rma.uni(yi=ES, vi=var, data=data, method=method, weighted=T)
  SE <- ma$se # Standard error

  # Minimum effect size required for statistical significance ES.alpha <-
  qnorm(1 - alpha) * SE

  # Final calculation of power
  power <- 1 - pnorm((ES.alpha - ES)/SE)
```

```

# Return the function
print(data.frame(ES = ES,
                 Power = power))
}

## -CI range-          Function for calculating the width of the CI for the
combined TE estimate ##

# Arguments #
# -ES-              Column of a data frame containing the effect sizes from each
study (For metafor)
# -var-            Column of a data frame containing the variances for each study
(For metafor)
# -data-          Data frame containing the data (effect sizes and variances) (For
metafor)
# -method-        Method for estimating tau^2 (For metafor). The most common codes
are (run ?rma.uni for all codes):
# -"FE"-          (Fixed effect. Tau^2 is not estimated)
# -"REML"- (Maximum Likelihood with Constraints). Default # -alpha-
Significance level. Default = 0.05

range.CI <- function(ES, var, data, method="REML", alpha=0.05) {

# Calculate the standard error
ma <- rma.uni(yi=ES, vi=var, data=data, method=method, weighted=T)
SE <- ma$se # Standard error value

# Calculate the range
range <- 2 * SE * abs(qnorm(1 - alpha/2))

# Return the function print(range)
}

## -studies.power-    Function for the prospective calculation of the number of
studies/subjects needed to achieve a given power ##

# Arguments #
# -ES-              Column of a data frame containing the effect sizes for each
study (For metafor)
# -var-            Column of a data frame containing the variances for each study
(For metafor)

```



```

# -data-          Data frame containing the data (effect sizes and variances) (For
metafor)
# -method-        Tau^2 estimation method (For metafor). The most common codes are
(run ?rma.uni for all codes):
# -"FE" -         (Fixed effect. Tau^2 is not estimated)
# -"REML"-        (Restricted Maximum Likelihood). Default
# -theta-         Alternative parametric TE. By default, the power is calculated for the
classic cutoff points proposed by Cohen (0.2, 0.5, 0.8), and these values are
always offered first.
# -alpha-         Significance level. Default = 0.05 # -power-
Desired power value.
# -plot.limit-    Total number of studies to be estimated for the plot. Default = 50

studies.power <- function(ES, var, data, method="REML", theta=NULL, alpha=0.05,
power, plot.limit=50) {

  # Vector of sorted alternative effect sizes: es <-
sort(c(theta, 0.2, 0.5, 0.8))
  es <- es[!duplicated(es)] # Remove duplicates, if any

  # Calculation of the original study data
  ma <- rma.uni(yi=ES, vi=var, data=data, method=method, weighted=T) SE_k
  <- ma$se # Original standard error

  W_k <- 1/SE_k^2 # Total weight of the original studies

  w.mean_k <- W_k/length(data$ES) # Mean weight of the original studies

  # Calculation of the standard error for the k+j
  studies SE_kj <- es/(qnorm(1 - alpha) - qnorm(1 -
power))

  # Calculation of the weight of the k+j
  studies W_kj <- 1/SE_kj^2

  # Calculate the weight of the j studies
  W_j <- W_kj - W_k

  # Calculate the variance for the j studies
  var_j <- 1/W_j

  # Calculating the estimated sample size (only if method = EF)
  if (method == "FE") {
    Subjects <- 1:length(es)

```

```

    for (i in 1:length(Subjets)) {
      Subjets[i] <- ceiling((4/var_j[i]) + ((2 * es[i]^2)/(4 * var_j[i])))
    }
  }

# Calculation of estimated studies
Studies <- ceiling(W_j/w.mean_k)

# Calculating study-by-study power (for the graph) # Data frame
to hold the data
data.plot <- data.frame(j = 0:(plot.limit-1))

for (i in 1:length(es)) {
  # Vector containing the power values for a given TE
  c <- 1 - pnorm(((qnorm(1 - alpha) * SE_k) - es[i])/SE_k)

  # Loop control mechanism meter <-
  0

  # Loop
  repeat {
    # Update the counter meter <-
    meter + 1

    # Recalculation of the standard error with the k+j
    studies Wkj <- (1/SE_k^2) + (meter * w.mean_k)
    SEkj <- sqrt(1/Wkj)

    # Recalculate the power
    powerkj <- 1 - pnorm(((qnorm(1 - alpha) * SEkj) - es[i])/SEkj)

    # Assign power to the vector c[meter] <-
    powerkj

    # Close the loop
    if (meter == plot.limit)
      break
  }

  # Append the new vector to the data frame
  data.plot <- cbind(data.plot, c)
}

```

```

# Rename data frame columns
cnames <- c("j", paste("ES=", es, sep=""))
colnames(data.plot) <- cnames

# Creating the plot (First line) plot(x=data.plot[,1],
y=data.plot[,2],
    type="l", lwd = 2,
    xlab = "Number of studies", ylab = "Power",
    ylim=c(min(data.plot[,2]),max(data.plot[,ncol(data.plot)])), col=2)
# Next lines
for (i in 3:ncol(data.plot)) {
    lines(x=data.plot[,1], y=data.plot[,i], col=i, lwd=2, type="l")
}
# Vertical lines for the number of studies that reach the specified power
for (i in 1:length(cnames)) {
    abline(v=Studies[i], col=(i+1))
}

# Legend legend("bottomright",
    inset=0.03,
    legend=cnames[-1],
    col=c(2:ncol(data.plot)),
    lwd=2)

# Function return (for method = FE) if
(method == "FE"){
    results <- list(Subjets = data.frame(ES = es,
                                         Subjets = Subjets),
                   Studies = data.frame(ES = es,
                                         Studies = Studies),
                   Data.plot = data.plot)
} else {
    results <- list(Studies = data.frame(ES = es,
                                         Studies = Studies),
                   Data.plot = data.plot)
}

# Negative values and NAs are coerced to 0 for (i in
1:length(names(results))){
    results[[i]][results[[i]] < 0] <- 0
    results[[i]][is.na(results[[i]])] <- 0
}

```

```

    }

    print(results)
}

## -studies.range- Function for the prospective calculation of the number of
studies/subjects needed to achieve a given CI width ##

# Arguments #
# -ES-          Column of a data frame containing the effect sizes for each
study (For metafor)
# -var-         Column of a data frame containing the variances for each study
(For metafor)
# -data-        Data frame containing the data (effect sizes and variances) (For
metafor)
# -method-      Method for estimating tau^2 (For metafor). The most common codes
are (run ?rma.uni for all codes):
# -"FE"-        (Fixed effect. Tau^2 is not estimated)
# -"REML"-      (Restricted Maximum Likelihood). Default # -alpha-
Significance level. Default = 0.05
# -range-       Desired width of the CI.
# -theta-       Alternative parametric TE (Only required if method="FE" and
metric="SMD". By default, power is calculated for the classic cut-off points
proposed by Cohen (0.2, 0.5, 0.8), and these higher values are always offered.
# -plot.limit-  Total number of studies to be estimated for the plot. Default = 50

studies.range <- function(ES, var, data, method="REML", alpha=0.05, range,
theta=NULL, plot.limit=50) {

  # Vector of sorted alternative effect sizes (Used only if method="FE" and
metric="SMD")
  es <- sort(c(theta, 0.2, 0.5, 0.8))
  es <- es[!duplicated(es)] # Remove duplicates if any

  # Calculation of the original study data
  ma <- rma.uni(yi=ES, vi=var, data=data, method=method, weighted=T) SE_k
  <- ma$se # Original standard error

  W_k <- 1/SE_k^2 # Total weight of the original studies

  w.mean_k <- W_k/length(data$ES) # Mean weight of the original studies

```

```

# Calculation of the standard error for the k+j
studies SE_kj <- range/(2 * abs(qnorm(1 - alpha/2)))

# Calculation of the weight of the k+j
studies W_kj <- 1/SE_kj^2

# Calculate the weight of the j studies
W_j <- W_kj - W_k

# Calculation of the variance of the j
studies var_j <- 1/W_j

# Calculate the estimated sample size (only for method="FE") if
(method == "FE") {
  Subjects <- 1:length(es)
  for (i in 1:length(Subjects)) {
    Subjects[i] <- ceiling((4/var_j) + ((2 * es[i]^2)/(4 * var_j)))
  }
}

# Calculation of estimated studies
Studies <- ceiling(W_j/w.mean_k)

# Calculate the range study by study (for the graph)
# Data frame containing the information needed for the graph (number of studies
added and the corresponding range)
data.plot <- data.frame(j = 0,
                        range = range <- 2 * SE_k * abs(qnorm(1 - alpha/2))
)

# Loop control mechanism meter <- 0

# Loop repeat
{
  # Update the counter meter <-
meter + 1

  # Recalculation of the standard error with the k+j
studies Wkj <- (1/SE_k^2) + (meter * w.mean_k)
SEkj <- sqrt(1/Wkj)

```

```

# Recalculation of the amplitude
rangekj <- 2 * SEkj * abs(qnorm(1 - alpha/2))

# Assign the new data to the data frame data.plot <-
rbind(data.plot, c(meter, rangekj))

# End of loop
if (meter == (plot.limit - 1))
  break
}

# Plot
plot(data.plot, type="l", xlab = "Number of studies", ylab = "Amplitude", lwd
= 2)
abline(v = Studies)

# Function return (for method = FE) if
(method == "FE"){
  if (metric == "SMD"){
    results <- list(Subjects = data.frame(ES = es,
                                           Subjects = Subjects),
                  Studies = Studies,
                  Data.plot = data.plot)
  } else if (metric == "PC"){
    results <- list(Subjects = Subjects,
                  Studies = Studies,
                  Data.plot = data.plot)
  }
}

} else {
  results <- list(Studies = Studies,
                Data.plot = data.plot)
}

# Negative values and NAs are coerced to 0 for (i in
1:length(names(results))){
  results[[i]][results[[i]] < 0] <- 0
  results[[i]][is.na(results[[i]])] <- 0
}

print(results)
}

```

8.2. Appendix 2. Data from the meta-analysis by Velthorst et al. (2015)

Table 2

Data from the meta-analysis by Velthorst et al. (2015) required for the example

n	EN	var
1	0.389	0.027889
2	1.761	0.229441
3	0.364	0.044521
4	1.235	0.123904
5	0.7	0.112896
6	0.372	0.095481
7	0.445	0.081225
8	-0.425	0.046225
9	0.563	0.044944
10	0.013	0.135424
11	-0.495	0.072361
12	-0.313	0.035344
13	-0.044	0.056169
14	0.375	0.097969
15	-0.121	0.033124
16	-0.112	0.018225
17	0.214	0.057121
18	0.188	0.063001
19	-0.096	0.060025
20	-0.123	0.042849
21	0.12	0.0529
22	0.208	0.049284
23	-0.008	0.126736
24	0.066	0.085849
25	0.414	0.097969
26	-0.386	0.062001
27	-0.422	0.039204
28	-0.126	0.051984

8.3. Appendix 3. Output of the “studies.power” function (\$Data.plot)

```
$Data.plot
      j      ES=0.2      ES=0.5      ES=0.8
1      0      0.8511480      0.9999998      1
2      1      0.8615460      0.9999999      1
3      2      0.8712782      0.9999999      1
4      3      0.8803810      1.0000000      1
5      4      0.8888896      1.0000000      1
6      5      0.8968378      1.0000000      1
7      6      0.9042578      1.0000000      1
8      7      0.9111807      1.0000000      1
9      8      0.9176361      1.0000000      1
10     9      0.9236521      1.0000000      1
11    10      0.9292558      1.0000000      1
12    11      0.9344727      1.0000000      1
13    12      0.9393269      1.0000000      1
14    13      0.9438415      1.0000000      1
15    14      0.9480382      1.0000000      1
16    15      0.9519376      1.0000000      1
17    16      0.9555592      1.0000000      1
18    17      0.9589212      1.0000000      1
19    18      0.9620410      1.0000000      1
20    19      0.9649348      1.0000000      1
21    20      0.9676178      1.0000000      1
22    21      0.9701044      1.0000000      1
23    22      0.9724082      1.0000000      1
24    23      0.9745417      1.0000000      1
25    24      0.9765169      1.0000000      1
26    25      0.9783448      1.0000000      1
27    26      0.9800358      1.0000000      1
28    27      0.9815996      1.0000000      1
29    28      0.9830453      1.0000000      1
30    29      0.9843815      1.0000000      1
31    30      0.9856159      1.0000000      1
32    31      0.9867561      1.0000000      1
33    32      0.9878088      1.0000000      1
34    33      0.9887805      1.0000000      1
35    34      0.9896771      1.0000000      1
36    35      0.9905043      1.0000000      1
37    36      0.9912672      1.0000000      1
38    37      0.9919705      1.0000000      1
39    38      0.9926188      1.0000000      1
40    39      0.9932162      1.0000000      1
41    40      0.9937666      1.0000000      1
42    41      0.9942735      1.0000000      1
43    42      0.9947403      1.0000000      1
44    43      0.9951700      1.0000000      1
45    44      0.9955655      1.0000000      1
46    45      0.9959293      1.0000000      1
47    46      0.9962641      1.0000000      1
48    47      0.9965719      1.0000000      1
49    48      0.9968550      1.0000000      1
50    49      0.9971152      1.0000000      1
```


8.4. Appendix 4. Output of the “studies.range” function (\$Data.plot)

```
$Data.plot
      j      power
1      0 0.2970202
2      1 0.2918542
3      2 0.2869487
4      3 0.2822826
5      4 0.2778369
6      5 0.2735949
7      6 0.2695414
8      7 0.2656629
9      8 0.2619472
10     9 0.2583831
11    10 0.2549606
12    11 0.2516707
13    12 0.2485049
14    13 0.2454556
15    14 0.2425159
16    15 0.2396794
17    16 0.2369401
18    17 0.2342927
19    18 0.2317320
20    19 0.2292535
21    20 0.2268529
22    21 0.2245261
23    22 0.2222695
24    23 0.2200796
25    24 0.2179532
26    25 0.2158873
27    26 0.2138790
28    27 0.2119257
29    28 0.2100250
30    29 0.2081745
31    30 0.2063721
32    31 0.2046157
33    32 0.2029034
34    33 0.2012334
35    34 0.1996039
36    35 0.1980134
37    36 0.1964604
38    37 0.1949433
39    38 0.1934608
40    39 0.1920116
41    40 0.1905946
42    41 0.1892084
43    42 0.1878520
44    43 0.1865245
45    44 0.1852246
46    45 0.1839516
47    46 0.1827044
48    47 0.1814823
49    48 0.1802844
50    49 0.1791099
51    50 0.1779581
52    51 0.1768281
53    52 0.1757195
54    53 0.1746314
55    54 0.1735633
56    55 0.1725146
57    56 0.1714847
58    57 0.1704730
59    58 0.1694789
60    59 0.1685021
```

61	60	0.1675420
62	61	0.1665981
63	62	0.1656699
64	63	0.1647571
65	64	0.1638593
66	65	0.1629759
67	66	0.1621067
68	67	0.1612513
69	68	0.1604092
70	69	0.1595802
71	70	0.1587640
72	71	0.1579601
73	72	0.1571683
74	73	0.1563883
75	74	0.1556198
76	75	0.1548625
77	76	0.1541162
78	77	0.1533806
79	78	0.1526553
80	79	0.1519403
81	80	0.1512353
82	81	0.1505399
83	82	0.1498541
84	83	0.1491775
85	84	0.1485101
86	85	0.1478515
87	86	0.1472016
88	87	0.1465602
89	88	0.1459271
90	89	0.1453021
91	90	0.1446851
92	91	0.1440759
93	92	0.1434744
94	93	0.1428803
95	94	0.1422935
96	95	0.1417139
97	96	0.1411413
98	97	0.1405756
99	98	0.1400166
100	99	0.1394643